

PHOBIA-FL: Personalized Healthcare using Ordinality, Bayesian Learning and Intersection Similarity Metric based Aggregation in Federated Learning

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Abstract—Medical cohorts, i.e., patient datasets from different hospitals or institutions, are highly heterogeneous and sensitive, making conventional centralized learning approaches challenging. Federated learning (FL) overcomes this limitation by training models on the local provider infrastructure without ever exposing raw records, a critical requirement in healthcare. However, the highly heterogeneous distribution of medical cohorts across providers limits the effectiveness of traditional FL approaches, thereby necessitating more advanced solutions. Recently, Bayesian Neural Networks (BNNs) have gained popularity in healthcare data analytics due to their ability to estimate uncertainty and quantify prediction confidence. In addition, BNNs can achieve a balance in personalization by prioritizing the prior model or evidence based on the quality of the local cohort. This work proposes a novel framework, called PHOBIA-FL, for healthcare providers to train a personalized collaborative model using BNN and ISM (intersection similarity metric)-based aggregation in FL, to handle highly heterogeneous medical cohorts. The framework supports both the provider-side and orchestrator-side components with minimal changes from existing FL protocols. Using BNN-based model personalization, high performance is achieved in global test records, even for providers with limited cohort size. Experiments with highly heterogeneous record distributions in sparsely sampled cohorts also showcase an improved performance and uncertainty estimation for PHOBIA-FL.

Index Terms—Federated Learning, Bayesian Neural Networks, Personalization in FL

I. INTRODUCTION

Safeguarding the privacy of patient medical records is an essential consideration for most healthcare providers. Various legislations enforce strict medical record privacy restricting medical cohort sharing among providers even for data analytics. This makes collaborative learning via pooling of medical cohorts across providers infeasible. In light of these restrictions, *Federated Learning* (FL) [1] helps train a collaborative model for medical cohort analytics without the records leaving the provider. Naturally, FL is increasingly leveraged in sensitive medical environments [2] for collaborative learning. In FL, the providers train on their local cohorts and only communicate their models intermittently to an orchestrator. The orchestrator is in charge of aggregating the submitted models and broadcasting the aggregated model back to the providers. The aggregated model (also referred to as the global model) is thus able to perform the desired analytics on the providers' local cohorts. This eliminates the need to share cohorts among the providers or the orchestrator thereby remaining compliant with the cohort privacy obligations.

However, traditional FL methods are inefficient in scenarios where the provider data is highly heterogeneous. Especially, among providers with different record collection capacities, there is a need for increased personalization in the learning process. To this end, Bayesian Neural Networks (BNNs) are increasingly used in healthcare data analytics to provide balanced personalization between the cohort evidence and the known prior. This allows a pre-determined model to be established as a shared initialization that each provider can adapt to its local cohort distribution. Thus, BNNs are able to satisfy the personal desires of the providers while providing prior initialization to providers with insufficient cohort diversity.

In this work, we introduce a novel FL framework called PHOBIA-FL that integrates Bayesian Neural Networks with FL to enable providers to obtain global priors from an orchestrator. PHOBIA-FL balances the relative influence of the global prior using the provider cohort diversity. With the help of Intersection Similarity Metric (ISM), this method allows providers with poor cohort quality to gain more from the collaborative model whereas allowing the providers with better cohort quality to rely more on local training. Additionally, PHOBIA-FL extends the personalization to aggregation phase by calibrating the collaborative model towards a consensus that respects each provider's cohort characteristics.

The rest of the manuscript is organized as follows: Section II provides essential preliminaries for FL and BNNs and highlights their utilities in healthcare data analytics. Section III proposes the PHOBIA-FL framework along with the corresponding provider-side and orchestrator-side algorithms. Section IV validates the superiority of PHOBIA-FL framework using medical datasets under highly heterogeneous environments. Finally, Section V concludes the paper and suggests directions for future research in the area.

II. BACKGROUND AND RELATED WORK

This section provides background on FL setup and provides the preliminary on BNNs. It also discusses existing literature that uses the methodologies for healthcare data analytics.

A. FL Objective

Using FL, we aim to train a collaborative model $W_C \in \mathbb{R}^d$ by aggregating local models w_i across a set of providers, $\mathcal{P}$.

$$W_C^{T+1} = \frac{1}{N} \sum_{i=1}^N w_i^T, \forall p_i \in \mathcal{P} \quad (1)$$

where $N = |\mathcal{P}|$ is the number of providers and T is the current round of training. At the beginning of a round T , the collaborative model W_C^T is distributed to all providers by an orchestrator. Afterwards, a provider $p_i \in \mathcal{P}$ trains their local model w_i^T on their locally stored dataset $\mathcal{D}_i = \{X_i, y_i\}$, where X_i and y_i are the collected data samples and labels, respectively. The local objective for provider p_i is defined as

$$L_i(w_i^T) = \frac{1}{|\mathcal{D}_i|} \sum_{j \in |\mathcal{D}_i|} \nabla l(W_C^T(X_i^j) = y_i^j), \quad (2)$$

where l is p_i 's loss on the local task and W_C^T is the initial model communicated by the orchestrator. w_i^T is then updated using ∇L_i using gradient descent and communicated back to the aggregator. After aggregating the set of w_i according to Eq. (1), the collaborative model W_C^{T+1} is broadcasted back to the participants for the next FL round. After multiple such iterations, we obtain the final collaborative model W_C^* that is convergent on diverse data without any provider data sharing.

FL's privacy-preserving framework allows providers to train shared models without exchanging raw data ensuring compliance and better generalization for diverse cohorts [2]. Personalized aggregation and fairness-aware weighting further address heterogeneous datasets to produce robust site-specific models. Extensions like MetaFed's cyclic knowledge distillation [3], pFedCAS's link-quality sparsification [4], and metadata-augmented bayesian classifiers [5] then fine-tune models to local distributions, enhancing diagnostic accuracy.

B. Bayesian Neural Networks

For a standard Bayesian Neural Network(BNN) with variational inference [6], we have an optimization problem that aims at finding the closest distribution $w(v)$ to the posterior distribution $\pi(v|\mathcal{D})$ in the variational family of distributions $\mathcal{V}$ given evidence $\mathcal{D}$. Mathematically, we define this as

$$\min_{w(v) \in \mathcal{V}} KL(w(v)||\pi(v|\mathcal{D})) \quad (3)$$

where KL is the KL-Divergence between the distributions. Through Bayes theorem, we establish $\pi(v|\mathcal{D}) \propto \pi(v)p_v(\mathcal{D})$, where $\pi(v)$ denotes the prior distribution and $p_v(\mathcal{D})$ denotes the likelihood. Thus, the optimization problem is given as

$$\min_{w(v) \in \mathcal{V}} -\mathbb{E}_{w(v)}[p_v(\mathcal{D})] + KL(w(v)||\pi(v)) \quad (4)$$

where the first term denotes the loss on the dataset $\mathcal{D}$ and the second term is the divergence from prior distribution. The optimization is solved using a first-order-stochastic gradient descent algorithm for Eq. (4) using variational inference [6]. The KL divergence in Eq. (3) and Eq. (4) refers to the difference between the variational distribution and the true posterior. The random variable v_m in v is thus parameterized as

$$v_m = \mu_m + \log(1 + \exp(\rho_m)) \cdot \epsilon_m; \epsilon_m \sim \mathcal{N}(0, 1) \quad (5)$$

This ensures the model parameters are non-deterministic therefore providing non-deterministic outputs for fixed inputs.

BNN's non-deterministic outputs capture parameter uncertainty by maintaining full posterior distributions over weights,

yielding calibrated confidence estimates crucial for risk-sensitive clinical decision-making [6]. By incorporating informative priors, BNNs regularize learning on small, heterogeneous medical cohorts, mitigating overfitting and enhancing generalization [7]. Their probabilistic outputs also enable the automated deferral of low-confidence cases to clinicians and integrate seamlessly with federated frameworks for personalized, privacy-preserving diagnostic models [8].

III. PROPOSED METHODOLOGY PHOBIA-FL

PHOBIA-FL leverages the collaborative model communicated in FL as an established prior for the BNN optimization. Additionally, to balance the degree of personalization and the influence of the collaborative model, the algorithm weighs the contribution of the collaborative model based on the diversity of local data. The providers' contribution to the collaborative model uses the ordinality of the providers' annotations in a power-of-choice strategy. The framework ensures that providers with insufficient records for training maximize their benefits from global model and providers with a better corpus are able to contribute maximally to the global model.

A. Bayesian Neural Networks in Federated Learning

In PHOBIA-FL, we provide a Bayesian Neural Network to the providers for training on the localized cohorts. We name this model as the "Personalized Model" w_i for the provider p_i . The "Collaborative Model" W_C^T is broadcasted at the beginning of the training round T to all $p_i \in \mathcal{P}$. Unlike regular FL, where the submitted W_C^T is substitutes an initial model, PHOBIA-FL providers only use W_C^T as the estimate for the prior model $\pi(w)$ in Eq. (4). We omit T from the succeeding notations for simplicity. Thus the objective for the provider p_i transforms to (i) minimize the local training loss on the annotated cohorts (ii) minimize the divergence from the collaborative global model. We describe both objectives as

a) *Loss minimization on the dataset:* To minimize the training loss, provider p_i samples mini-batches on the local cohorts. Afterwards, p_i performs C Monte-Carlo forward passes on the cohort and computes the loss. We name this as the local loss $L(w_i(v), \mathcal{D}_i)$ which is described as

$$L(w_i(v), \mathcal{D}_i) = 1/C \times \sum_{c=1}^C l(w_i(\{\mu, \rho, \epsilon_c\}, X_i) = y_i) \quad (6)$$

Here $w_i \sim \mathcal{W}$ is parameterized using the random vector $v \in \mathbb{R}^d$ such that $w_{i(v_m)}; \forall m = \{1, \dots, d\}$ is sampled using Eq. 5. Thus, the weights of w_i represent a vector where the elements are from a normal distribution $\mathcal{N}(\mu_m, \log^2(1 + \exp(\rho_m)))$; $\forall m$. This allows non-deterministic outputs for deterministic inputs. The loss values are calculated and averaged across C monte-carlo samples to obtain the mini-batch loss.

b) *Minimizing Divergence from the central model:* The model W_C is substituted as the prior model in Eq. (4) to find the closest distribution in $\mathcal{W}$ for w_i that minimizes the divergence with W_C^T . We achieve this by minimizing the KL divergence between w_i and W_C^T as:

$$\begin{aligned}
KL(w_i(v)||W_C(v)) &= \sum_{m=1}^d KL(w_i(v_m)||W_C(v_m)) \\
&= \sum_{m=1}^d KL(\mathcal{N}(\mu_{i,m}, \sigma_{i,m}^2)||\mathcal{N}(\mu_{C,m}, \sigma_{C,m}^2)) \\
&= \frac{1}{2} \sum_{m=1}^d [\log(\frac{\sigma_{C,m}^2}{\sigma_{i,m}^2}) + \frac{\sigma_{i,m}^2 + (\mu_{i,m} - \mu_{C,m})^2}{\sigma_{C,m}^2} - 1]
\end{aligned} \quad (7)$$

where v is the random vector as defined in Eq. (5).

In order to update w_i , we perform backpropagation on v to find the corresponding μ_m and ρ_m that minimize the sum of losses in Eqs. (6) and (7). However, prioritizing the minimization of Eq. (7) leads to w_i approximating W_C^T which is suitable for providers with limited cohort size but undesirable for providers with larger training cohorts. Similarly, for prioritizing Eq. (6) is beneficial to providers with larger training cohorts but detrimental otherwise. Thus, to ensure balanced personalization, we introduce a hyperparameter ζ inspired by the modification in β -VAE [9]. Thus the cumulative loss objectives for provider p_i is given as

$$L(p_i) = L(w_i(v), \mathcal{D}_i) + \zeta KL(w_i(v)||W_C(v)) \quad (8)$$

the obtained $\nabla_v L(p_i)$ then updates w_i .

B. Balancing Personalization using Cohort Diversity

Identifying the optimum ζ in Eq. (8) is essential for achieving the desired level of personalization. Inspired by [10], we leverage the intersection similarity metric (ISM) for representing the distribution of a provider's cohort as a set of annotation-count pairs and assigning the optimal ζ value to the provider. We outline the corresponding steps as follows

a) *Formation of Annotation Distribution Set:* Then, for each provider $p_i \in \mathcal{P}$ we count the number of records n_ℓ belonging to each diagnosis category $\ell \in L$ where L is the total number of diagnosis in p_i 's cohort. The resulting collection of (ℓ, n_ℓ) pairs contains the distribution of categories in that provider's p_i 's records such that $Q_i = \{(\ell, n_\ell) \forall \ell \in L\}$. For each $Q_i^l \in Q_i$, we construct a set as

$$\mathcal{Z}_i = \bigcup_{l=1}^L \bigcup_{n=1}^{n_\ell} (l \times Q_{\max} + n) \quad (9)$$

where $\mathcal{Z}_i$ contains the encoded value for the annotation distribution for p_i which provides a way to give each sample in a p_i 's cohort a distinct, non-overlapping index. It is worth mentioning that $Q_{\max}$ value simply needs to exceed the largest per label sample count. Afterwards, the $\mathcal{Z}_i \forall p_i$ is communicated to the orchestrator.

b) *Formation of Intersection Similarity Metric:* We define $ISM(i, j)$ for p_i with respect to p_j as

$$ISM(i, j) = \frac{|\mathcal{Z}_i \cap \mathcal{Z}_j|}{|\mathcal{Z}_j|}; ISM(i, j) \in [0, 1]. \quad (10)$$

which quantifies the fraction of p_i 's indexed annotations that p_j also possesses. $ISM(i, j) = 1$ means that p_j covers all

Algorithm 1: PHOBIA-FL: Provider-Side

```

1 function provider_training( $W_C, \zeta_i$ ):
2   for  $\mathcal{D}_i^j \in \mathcal{D}_i$  do // mini-batches
3      $L(w_i(v), \mathcal{D}_i^j) \leftarrow 0$ 
4     for  $c \in [1, \dots, C]$  do // Monte-Carlo Samples
5        $v_m \leftarrow \mu_m + \log(1 + \exp(\rho_m)) \cdot \epsilon_m$ 
6        $L(w_i(v), \mathcal{D}_i^j) \leftarrow$ 
         $L(w_i(v), \mathcal{D}_i^j) + \frac{1}{C} l(w_i(\{\mu, \rho, \epsilon_c\}, X_i) = y_i)$ 
7        $L(p_i) \leftarrow L(w_i(v), \mathcal{D}_i^j) + \zeta_i KL(w_i(v)||W_C(v))$ 
8        $w_i(v) \leftarrow w_i(v) - GD(\nabla_{w_i(v)} L(p_i))$ 
9   return  $w_i$ 

1 function annotation_distribution( $Q_{\max}$ ):
2    $\mathcal{Z} \leftarrow \emptyset$ 
3   for  $\ell \in \text{unique}(y_i)$  do // for all labels
4      $n_\ell \leftarrow |\{\mathcal{D}_i : y_i = \ell\}|$ 
5      $\mathcal{Z} \leftarrow \mathcal{Z} \cup \{\ell Q_{\max} + 1, \dots, \ell Q_{\max} + n_\ell\}$ 
6   return  $\mathcal{Z}$ 

1 function local_loss( $W_C$ ):
2    $loss \leftarrow 0$ 
3   for  $\mathcal{D}_i^j \in \mathcal{D}_i$  do // mini-batches
4      $v_m \leftarrow \mu_m + \log(1 + \exp(\rho_m)) \cdot \epsilon_m$ 
5      $loss \leftarrow loss + \frac{1}{C} l(W_C(\{\mu, \rho, \epsilon_c\}, X_i) = y_i)$ 
6    $loss \leftarrow loss / |\mathcal{D}_i^j|$ 
7   return  $loss$ 

```

of p_i 's cohort distribution, but not vice-versa as ISM is asymmetric. We define a provider p_i 's ISM vector as

$$ISM(i) = (ISM(i, 1), ISM(i, 2), \dots, ISM(i, N)) \quad (11)$$

c) *Diversity-Optimized Personalization:* The orchestrator collects the ISM vectors $ISM(i); \forall p_i \in \mathcal{P}$. For each of these obtained vectors, the orchestrator computes the sum of all ISM values in the submitted ISM vector as $ISM_\Sigma(i) = \sum_j ISM(i, j); \forall j \in \mathcal{P} \setminus p_i$.

For each provider p_i , the orchestrator then computes the corresponding ζ_i as $\zeta_i = N/ISM_\Sigma(i)$. For providers with no overlapping annotation count-pairs, the $ISM_\Sigma(i)$ value equals 0 which indicates the possession of annotations not present in the other providers' corpus. For those providers, we assign the value of $\zeta = N$ to balance the contribution of the collaborative model. We mention to the reader that a lower value of $ISM_\Sigma(i)$ indicates the distribution of the provider p_i 's annotations being non-overlapping from the other providers thereby indicating a lack of diversity in local records. As such, the corresponding ζ_i value is higher thereby allowing the personalized model w_i to learn more from the collaborative model W_C . Similarly, a higher $ISM_\Sigma(i)$ indicates the contrary.

C. Relative Ordinance based Aggregation

In order to maximize collaborative model performance, the provider models need to be aggregated based on the local cohort quality. Unlike traditional FL which leverages data proportion only [1], we leverage overall cohort quality of the provider by performing a forward pass on the provider data. To this end, PHOBIA-FL utilizes a three step algorithm for identifying the optimum provider weights for aggregation.

a) *Obtaining loss on the local dataset:* To identify providers with better training records, we sample N providers from $\mathcal{P}$ to populate a candidate set K . The probability for the selection of a provider p_i is given as $k_i = |\mathcal{D}_i| / \sum_{p_i} |\mathcal{D}_i|$; $\forall p_i$ thereby incentivizing providers with larger datasets. More importantly, this sampling is done without replacement therefore allowing multiple instances of a provider p_i in K . We consolidate the duplicates and provide the current collaborative model W_G to each provider $p_i \in K$. The provider p_i then performs a forward pass on the local cohorts and returns the corresponding loss value l_i . The returned loss values populate the loss array where an entry l_i corresponds to the loss value returned by p_i . The l_i values corresponding to $p_i \notin K$ are set to 0. The process is repeated for C Monte-Carlo samples to obtain different loss arrays.

b) *Ascertaining Ordinality of the Providers:* Upon obtaining the average of the losses obtained from the monte-carlo sampling, PHOBIA-FL sorts the obtained l_i values in ascending order $\forall p_i$. Afterwards, the ordinality of the losses are taken and the ω vector is updated with the same. Since the providers with the higher local loss indicate higher cohort quality [11], their personalized models contribute more to the aggregation. Thus, higher ordinality for a provider represents increased importance of their personalized model. We store the sum of these ordinalities in ω for the C monte-carlo samples.

c) *Normalizing and Clipping:* The obtained ω are normalized to have a max value of 1. Afterwards, the normalized values are passed through a logit function centered around 0.5 which helps to increase the divergence of the ω values [12]. The values are finally clipped to be in the range $[0, 1]$ to represent the fraction of contribution from the provider models.

After obtaining the ω values, the collaborative model is updated using Weighted Conflation (WC) [13] on w_i 's μ and σ values. WC effectively implements a weighted geometric mean in the log domain similar to log-linear pooling (LLP). In PHOBIA-FL, the weighted conflation (WC) rule incorporates the providers' contribution values ω into the aggregation as

$$\mu_{m,C} = \frac{1}{\tau} \sum_i \omega_i \cdot \mu_{m,i} / \sigma_{m,i}^2 \quad \sigma_{m,C}^2 = \max_i \omega_i / \tau \quad (12)$$

where $\tau = \sum_i \omega_i / \sigma_{m,i}^2$. Thus using Eq. (12), we obtain the posterior μ and σ values for the collaborative model W_C^T which then serve as the prior for the provider models for the $T+1$ round. Using WC ensures that the produced Gaussian's mean balances provider importance and variance is never larger than that of the most heavily weighted provider.

D. Overview of Algorithms

We outline the provider-side and the orchestrator-side setups for PHOBIA-FL in Algorithms 1 and 2 respectively. The orchestrator initiates the collaborative training process in `FL_training` (Algorithm 2 lines 1-10) for $T = T_{max}$ rounds. The ζ values are first obtained through `zeta_values` (Algorithm 2 lines 11- 20). The function obtains the individual providers' annotation distribution from `annotation_distribution` (Algorithm 1 lines 1-6).

Algorithm 2: PHOBIA-FL: Orchestrator-Side

Input: Initialized Collaborative Model W_C^0
Output: Converged Collaborative Model W_C^*

```

1 function FL_training( $W_C^0$ ):
2   for  $T \in \{0, \dots, T_{max}\}$  do //  $T_{max}$  FL rounds
3      $\zeta \leftarrow \text{zeta\_values}()$ 
4     for  $p_i \in \mathcal{P}$  do // Provider Side Training
5        $w_i \leftarrow \text{provider\_training}(W_C^T, \zeta_i)$ 
6      $\omega \leftarrow \text{aggregation\_weights}()$ 
7      $\mu_C \leftarrow \text{Conflation}(\omega_i, \mu_i)$ 
8      $\sigma_C \leftarrow \text{Conflation}(\omega_i, \sigma_i)$ 
9      $W_C^{T+1} \leftarrow W_C^T(\mu_C, \sigma_C)$ 
10    return  $W_C^* \leftarrow W_C^{T_{max}}$ 

11 function zeta_values():
12    $\mathcal{Z} \leftarrow \text{annotation\_distribution}(Q_{max})$ 
13    $\zeta_i \leftarrow 1 \forall p_i \in \mathcal{P}$ 
14   for  $p_i \in \mathcal{P}$  do
15      $ISM_{\Sigma}(i) \leftarrow 0$ 
16     for  $p_j \in \mathcal{P} \setminus p_i$  do
17        $ISM_{\Sigma}(i) \leftarrow |\mathcal{Z}_i \cap \mathcal{Z}_j| / |\mathcal{Z}_i|$ 
18     if  $ISM_{\Sigma}(i) > 0$  then
19        $\zeta_i \leftarrow 1 / ISM_{\Sigma}(i)$ 
20   return  $\zeta$ 

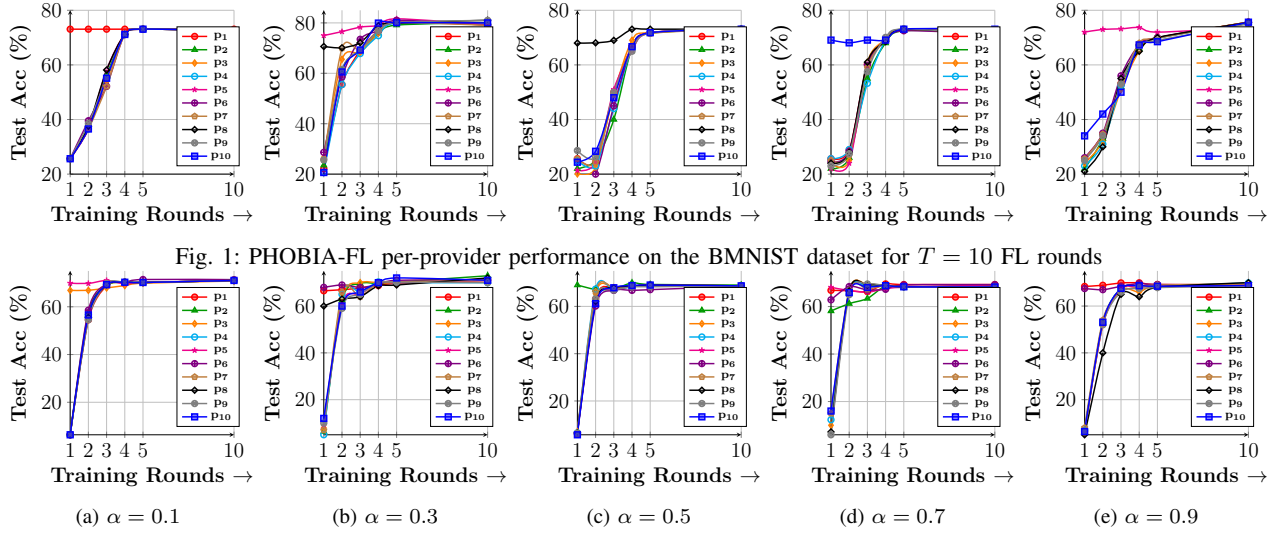
21 function aggregation_weights():
22    $\omega \leftarrow 0$ 
23   for  $c \in [1, \dots, C]$  do // Monte-Carlo Sampling
24      $l \leftarrow 0; \forall p_i \in \mathcal{P}$ 
25      $k \leftarrow |\mathcal{D}_i| / \sum_{p_i} |\mathcal{D}_i|$ 
26      $K \leftarrow N$   $p_i$  chosen w/o replacement with prob  $k_i$ 
27     for  $p_i \in \text{set}(K)$  do // chosen providers
28        $l \leftarrow l + \text{local\_loss}(W_C)$ 
29      $l \leftarrow \text{sort}(l/C)$ 
30      $\omega \leftarrow \omega + \text{Relative Ordinality}(l)$ 
31    $\omega \leftarrow \frac{\omega}{\max(\omega)}$ 
32    $\omega \leftarrow \log(\frac{\omega}{1-\omega} + 0.5)$ 
33    $\omega \leftarrow \min(1, \max(0, \omega))$ 
34   return  $\omega / \sum \omega$ 

```

Upon obtaining the corresponding annotation distribution set, the individual ζ_i values are calculated and broadcasted to the providers for training (Algorithm 1 lines 1- 9). The providers train their personalized models which are then submitted to the orchestrator which then determines the corresponding weightage given to each submitted provider model using `aggregation_weights` (Algorithm 2 lines 21- 34). The `aggregation_weights` function obtains the collaborative model loss on the provider's cohorts using `local_loss` (Algorithm 1 lines 1- 7). After averaging the losses across Monte Carlo samples, the orchestrator obtains the normalized ω_i values $\forall p_i$. These ω_i values are used to perform weighted conflation to obtain the posterior μ_C and σ_C values for the collaborative model for the next training round. This process repeats until the converged collaborative model is obtained.

IV. EXPERIMENTS AND RESULTS

We evaluate the performance of PHOBIA-FL framework on a standard image classification task using healthcare datasets BreastMNIST [14] and DermaMNIST [15]. We perform a

Fig. 2: PHOBIA-FL per-provider performance on the DMNIST dataset for $T = 10$ FL rounds

binary classification task on BreastMNIST dataset with 546 gray scale ultrasound image samples of size 28×28 . On DermaMNIST dataset, we classify colored dermatoscopic images of size $28 \times 28 \times 3$ into seven diagnostic categories using 1003 training image samples. For both datasets, we hold out a fraction of samples as test dataset for validation after training.

A. Experimental Setup

To perform these tasks, we use a pre-trained ResNet-18 deterministic network that is converted to a BNN using MOPED [16]. We perform experimental simulation across 10 providers communicating with a single orchestrator for 30 FL rounds to maximize performance on the image classification task. To simulate heterogeneity consistent with the real-world providers' cohorts, we distribute the training data in a dirichlet distribution [12]. We vary the parameter α in the dirichlet distribution in the range $\{0.9, 0.7, 0.5, 0.3, 0.1\}$ where $\alpha = 0.1$ induces extreme heterogeneity in the provider training cohorts. We also compare PHOBIA-FL against the existing state-of-the-art FL frameworks and aggregation algorithms like pFedBayes [7], Krum & mKrum [17], ProCFL [10], and pFedMe [18] to establish the superiority of PHOBIA-FL.

B. Evaluation Metrics

From the obtained results, we aim to answer the following questions (i) How do the local provider models perform on the global test dataset for the PHOBIA-FL algorithm in highly heterogeneous settings? (ii) How does PHOBIA-FL compare with existing state-of-the-art FL setups and aggregation algorithms in these heterogeneous settings. (iii) How effectively PHOBIA-FL capture epistemic and aleatoric uncertainties compared to the existing state-of-the-art?

To this end, we report the accuracies on the held-out test dataset for each of the simulated providers for the different heterogeneous settings. Additionally, we report the test accuracy for PHOBIA-FL along with the compared state-of-the-art for different heterogeneous settings. Finally, we also report the area-under-curve values obtained for various uncertainty thresholds for both the model and predictive uncertainty.

C. Results and Discussions

1) *Performance of PHOBIA-FL*: Figures 1 and 2 showcase the per-provider performance for PHOBIA-FL under varying levels of heterogeneity. We observe the effect of cohort heterogeneity in the first round where the performance is highly varied across the providers. Certain providers observe high accuracy in the first training round owing to better cohort quality. However, providers with poor cohort quality benefit from PHOBIA-FL's balanced personalization and quickly improve their performance. This is more evident in Figures 1 and 2a where only a single provider has access to a large cohort whereas the other providers are unable to obtain good local cohorts due to higher induced heterogeneity. However, after $T = 10$ rounds, all the providers converge to a common accuracy due to PHOBIA-FL allowing providers with reduced cohorts to benefit more from the collaborative model. Additionally, the collaborative model performance is not diminished during aggregation due to PHOBIA-FL providing increased importance to the models of providers with higher quality cohorts during aggregation.

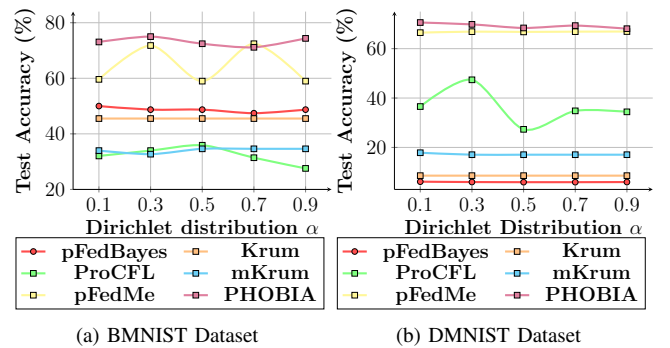


Fig. 3: Comparing PHOBIA-FL Test Accuracy with the existing state-of-the-art for different levels of heterogeneity

2) *Comparison with state-of-the-art*: Figures 3a and 3b compare PHOBIA-FL against the existing state-of-the-art for varied levels of data heterogeneity. For the both the datasets, we observe that the existing algorithms are super-

TABLE I: Area under Curve (AuC) values for varying levels of heterogeneity

Dataset	Algorithms	AuC for Model Uncertainty					AuC for Predictive Uncertainty				
		$\alpha = 0.1$	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 0.7$	$\alpha = 0.9$	$\alpha = 0.1$	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 0.7$	$\alpha = 0.9$
BMNIST	pFedMe	0.5848	0.647	0.5611	0.6229	0.5611	0.5136	0.5441	0.5322	0.5223	0.5322
	pFedBayes	0.5133	0.5357	0.5357	0.528	0.5357	0.4877	0.4925	0.4925	0.4787	0.4925
	Krum	0.3627	0.3627	0.3627	0.3627	0.3627	0.4716	0.4716	0.4716	0.4716	0.4716
	mKrum	0.3627	0.4037	0.4357	0.4357	0.4357	0.3225	0.3225	0.3225	0.3225	0.3225
	ProCFL	0.3396	0.3495	0.346	0.3216	0.3569	0.455	0.447	0.4579	0.4473	0.4425
	PHOBIA-FL	0.6434	0.4447	0.612	0.679	0.6631	0.5237	0.4736	0.5208	0.5399	0.5835
DMNIST	pFedMe	0.5705	0.5494	0.5787	0.5578	0.55	0.4844	0.493	0.4465	0.5008	0.4759
	pFedBayes	0.2842	0.2772	0.2392	0.2392	0.2772	0.4806	0.5388	0.4325	0.4325	0.4388
	Krum	0.2728	0.2728	0.2728	0.2728	0.2728	0.5264	0.5264	0.5264	0.5264	0.5264
	mKrum	0.2809	0.2707	0.2707	0.2707	0.2707	0.1774	0.2071	0.2071	0.2071	0.2071
	ProCFL	0.5001	0.596	0.3395	0.4542	0.4682	0.5101	0.428	0.5233	0.4729	0.5225
	PHOBIA-FL	0.5774	0.5736	0.6209	0.5682	0.6005	0.5275	0.5327	0.5643	0.54	0.5219

seded by PHOBIA-FL in performance. For the DMNIST dataset, pFedMe provides good performance comparable with PHOBIA-FL. However, its performance is highly inconsistent on the BMNIST Dataset whereas PHOBIA-FL shows consistent performance throughout. The other algorithms fail to compare with PHOBIA-FL in all cases showing poor performance. PHOBIA-FL performance shows marginal decline on $\alpha = 0.1$ and 0.7 for BMNIST and $\alpha = 0.5$ for DMNIST but still outperforms other algorithms consistently.

3) *Uncertainty quantification and analysis*: From the Area under curve values in Table I, we can observe the superiority of PHOBIA-FL against the compared algorithms. PHOBIA-FL provides the highest AuC value for Model Uncertainty estimation for the DMNIST dataset for all values of α . For the BMNIST dataset, PHOBIA-FL's uncertainty estimation is superseded only once by pFedMe. PHOBIA-FL's performance for predictive uncertainty estimation is comparable to the other algorithms. However, PHOBIA-FL is still the best performing algorithm for most values of α . pFedMe again shows robust performance comparable to PHOBIA-FL but fails to supersede it. pFedBayes also shows increased performance as compared to other algorithms as it also leverages BNNs but the performance is highly inconsistent on DMNIST.

V. CONCLUSION

In this work, we proposed the PHOBIA-FL framework, that is tailored for handling healthcare applications for critically heterogeneous cohorts. We integrate bayesian neural networks with a federated learning setup to enhance provider-side personalization and improve performance of providers with reduced cohorts. We leveraged an ISM-based similarity metric to enable diversity-aware personalization, with an ordinance-based aggregation on the orchestrator side. Experimental analysis on a two common medical datasets in highly heterogeneous settings validate the performance of PHOBIA-FL against the existing state-of-the-art FL frameworks. Future work will focus on improving aggregation robustness, expanding to more diverse datasets, and enhancing model reliability through uncertainty estimation that is crucial for trustworthy data analytics in healthcare domain.

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